Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	25 September 2026
Team ID	24bcs020@mscartsandscience-edu.in
Project Name	AI Fittrack
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2.

Example: High-Level End-to-End System Architecture (AI Fittrack)



Guidelines:
- Include all the process / technology blocks.
- Provide infrastructural details (e.g. AWS / GCP Cloud).
- Indicate external interfaces & AI model pipelines.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Cross-platform mobile app & web dashboard for interactive workout session tracking.	React Native, React.js, Tailwind CSS
2.	Application Logic-1	Backend service handling authentication, user profiles, and session scheduling.	Node.js / Express.js
3.	Application Logic-2	AI Processing logic for pose estimation and real-time exercise rep counter.	Python / FastAPI
4.	Application Logic-3	Notification & Real-time messaging service for exercise alerts.	Firebase Cloud Messaging (FCM)
5.	Database	Relational storage for user metadata, structured logs, and application configurations.	PostgreSQL
6.	Cloud Database	Managed Cloud Database instance with multi-AZ replication.	Amazon RDS for PostgreSQL

S.No	Component	Description	Technology
7.	File Storage	Object storage for raw workout video captures, model weights, and profile avatars.	AWS S3 / Firebase Storage
8.	External API-1	OAuth integration for user single sign-on authentication.	Google OAuth 2.0 / Facebook Login
9.	External API-2	Third-party fitness device data sync for heart rate and steps.	Google Fit API / Apple HealthKit
10.	Machine Learning Model	Computer vision and landmark pose estimation model for posture detection.	MediaPipe, OpenCV, TensorFlow Lite
11.	Infrastructure (Server / Cloud)	Containerized application deployment and cloud orchestrator hosting.	AWS EC2, Docker, Kubernetes, Cloudflare CDN

Table-2 : Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source libraries and frameworks used across the stack.	React Native, Node.js, OpenCV, MediaPipe, PostgreSQL
2.	Security Implementations	List all security & access controls implemented, use of encryption and firewalls.	AES-256 Encryption, JWT Auth, HTTPS/TLS 1.3, OWASP Compliance
3.	Scalable Architecture	Justify the scalability of architecture (3-tier, micro-services).	Microservices Architecture with Docker & Auto-scaling Groups
4.	Availability	Justify the availability of application (use of load balancers, distributed servers).	AWS ALB (Application Load Balancer) & Multi-region replication
5.	Performance	Design consideration for the performance of the application (caching, CDN).	Redis In-Memory Cache, TensorFlow Lite edge inference, Cloudflare CDN

References:

https://c4model.com/

https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture/

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d